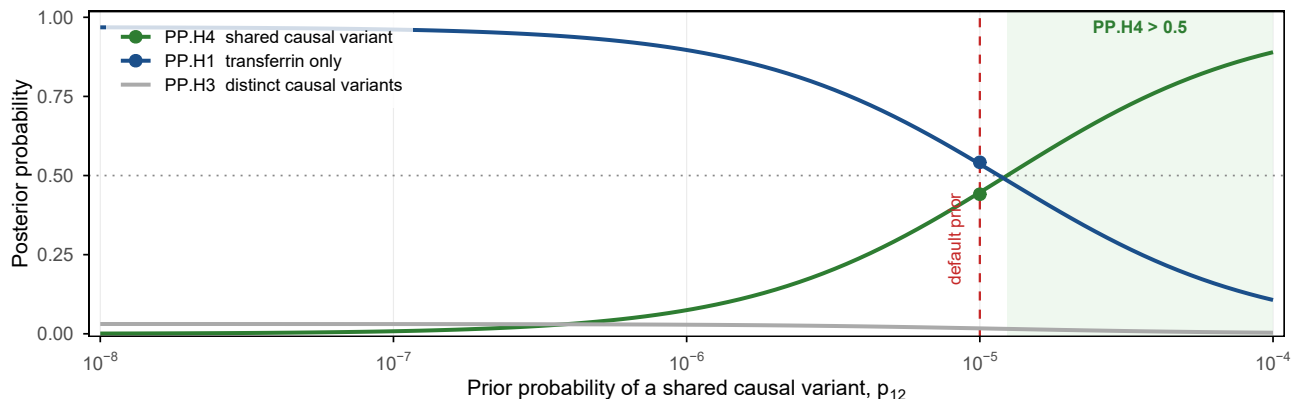
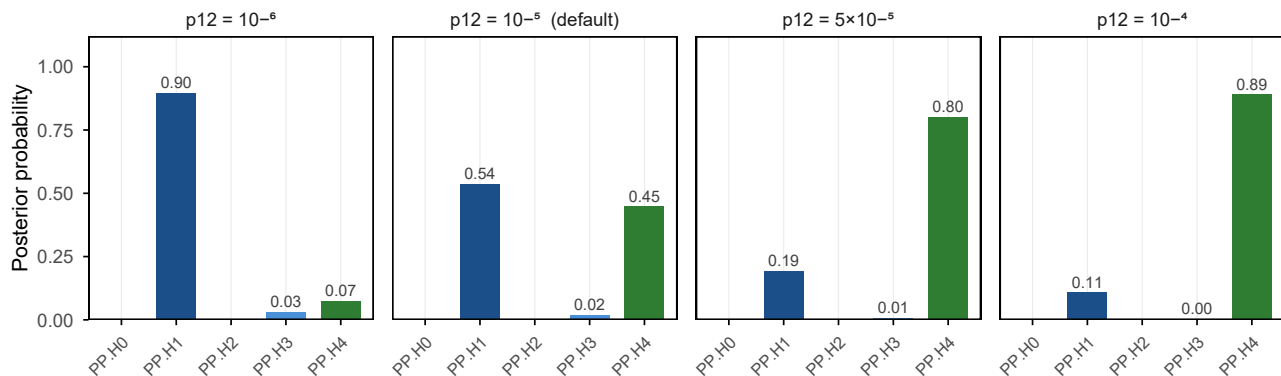


## A Posterior probabilities across the shared-variant prior



## B All five unconditional posteriors at selected priors



Sensitivity of the TF-locus colocalization to the shared-variant prior  $p_{12}$ . Panel A: coloc.abf re-run across  $p_{12}$ , with  $p_1 = p_2 = 1 \times 10^{-4}$  held fixed. At the default  $p_{12} = 1 \times 10^{-5}$  (dashed line) the transferrin-only hypothesis remains more probable than a shared causal variant (0.54 vs 0.45), and the conventional rule  $PP.H4 > 0.5$  is not met. The ordering is finely balanced: PP.H4 overtakes PP.H1 at  $p_{12} \approx 1.2 \times 10^{-5}$ , only about 20% above the default, and crosses 0.5 shortly after, at  $p_{12} \approx 1.24 \times 10^{-5}$  (shaded). Both values are interpolated, not grid points. Panel B: all five unconditional posteriors at selected priors. PP.H0 and PP.H2 are below 0.001 throughout. Numerical values are given in Supplementary Table S8.